

James Flemings

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Google Scholar: <https://scholar.google.com/citations?user=V5-ATAYAAAAJ&hl=en>

RESEARCH INTERESTS

My research broadly investigates privacy in language models. In particular, I'm interested in (1) improving privacy reasoning of language models; (2) principally understanding and measuring privacy leakage of language models (memorization, inference-time auditing); (3) controlling privacy leakage of language models (differential privacy).

EDUCATION

Ph.D. Computer Science

August 2022 – Current

University of Southern California

GPA: 3.83

Advisor: Murali Annavaram

B.S. Computer Science, Mathematics

August 2017 – May 2022

Minor: Computer Systems Engineering

University of Alaska Anchorage

GPA: 3.94

RESEARCH EXPERIENCE

Student Researcher

June 2025 – November 2025

Google

Mentor: Ren Yi; Federated Learning and Analytics Team

Topic: Personalizing Agents for Privacy Decisions

Research Scientist Intern

May 2024 – August 2024

TikTok

Mentor: Zafar Takhirov; Privacy Innovation Lab

Topic: Characterizing context privacy and hallucination in language models

Center for the Study of Language and Information Program

June 2022 – August 2022

Stanford University

Mentor: Christopher Potts

Topic: Building robust and interpretable AI with Interchange Intervention Training

Research Experiences for Undergraduates in Software Engineering

June 2021 – August 2021

Carnegie Mellon University

Mentor: Heather Miller; Composable Systems Lab

Topic: Developing a novel testing suite to benchmark Federated Learning algorithms

PUBLICATIONS

1. **J. Flemings**, R. Yi, O. Suci, K. Fawaz, M. Annavaram, M. Gruteser, "Personalizing Agent Privacy Decisions via Logical Entailment", 2025. Under Review.
2. J. Wei, A. Godbole, M. Khan, R. Wang, X. Zhu, **J. Flemings**, N. Kashyap, K. Gummadi, W. Neiswanger, R. Jia, "Hubble: a Model Suite to Advance the Study of LLM Memorization, 2025. Under Review.
3. M. Khan, A. Godbole, J. Wei, R. Wang, **J. Flemings**, K. Gummadi, W. Neiswanger, R. Jia, "Token-Smith: Streamlining Data Editing, Search, and Inspection for Large-Scale Language Model Training and Interpretability", In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*, 2025.
4. **J. Flemings**, H. Gan, H. Li, M. Razaviyayn, M. Annavaram, "Differentially Private In-context Learning via Sampling Few-shot Mixed with Zero-shot Outputs," 2025. Under Review.

5. A. Mulrooney, D. Gupta, **J. Flemings**, H. Zhang, M. Annavaram, M. Razaviyayn, X. Zhang, "DP-GRAPE: Memory-Efficient Differentially Private Training with Gradient Random Projection," 2025, Under Review.
6. **J. Flemings**, W. Zhang, B. Jiang, Z. Takhirov, M. Annavaram, "Estimating Privacy Leakage of Augmented Contextual Knowledge in Language Models," In *Proceedings of the 2025 Conference of the Association for Computational Linguistics*, 2025
7. **J. Flemings**, M. Annavaram, "Differentially Private Knowledge Distillation via Synthetic Text Generation," In *Findings of the 2024 Conference of the Association for Computational Linguistics*, 2024.
8. **J. Flemings**, M. Razaviyayn, M. Annavaram, "Differentially Private Next-Token Prediction of Large Language Models," In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics*, 2024.
9. **J. Flemings**, W. Zhang, B. Jiang, Z. Takhirov, M. Annavaram, "Characterizing Context Influence and Hallucination in Summarization," In *Towards Safe & Trustworthy Agents at Neurips*, 2024.
10. **J. Flemings**, M. Razaviyayn, M. Annavaram, "Adaptively Private Next-Token Prediction of Large Language Models," 2024, Under Review.
11. **J. Flemings**, M. Annavaram, "Differentially Private Knowledge Distillation via Synthetic Text Generation," In *PrivateNLP at ACL*, 2024.
12. **J. Flemings**, M. Razaviyayn, M. Annavaram, "Differentially Private Prediction of Large Language Models," In *The 5th Privacy-Preserving AI Workshop at AAAI*, 2024.

AWARDS

- NSF Graduate Research Fellowship April 2023
- USC-Meta Center Top up Fellowship August 2022
- Google CS Research Mentorship Program (CSRMP) Scholar September 2021

TALKS

1. "Differentially Private Prediction of Large Language Models." Tech Talk @ LinkedIn Research. July 2024.
2. "Differentially Private Prediction of Large Language Models." Tech Talk @ TikTok Privacy Innovation Lab. July 2024.
3. "Privacy in the Era of Large Language Models." Short Seminar @ USC Women in Science and Engineering (WISE). July 2024.
4. "Modular Monochromatic $(3, t)$ -colorings". 52nd Southeastern International Conference on Combinatorics, Graph Theory & Computing. Florida Atlantic University. 2021. Link: <https://www.youtube.com/watch?v=qciRVyWc90M>

PROFESSIONAL SERVICE

Reviewer

Neurips 2025 ICLR 2025 TMLR 2025 ACL 2025

Program Committee Member and Reviewer

<i>AAAI Workshop on Privacy Preserving Artificial Intelligence</i>	2024, 2025
<i>ACL Workshop on Large Language Model Memorization</i>	2025
<i>NAACL Workshop on Privacy in Natural Language Processing</i>	2025

Artifact Evaluation Committee Member

<i>Principles and Practice of Parallel Programming Conference</i>	2022
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TEACHING EXPERIENCE

Teaching Assistant

August 2019 – December 2022

University of Southern California

- **Courses:** CSCI 350: Introduction to Operating Systems

University of Alaska Anchorage

- **Courses:** CSCI 311 Data Structures and Algorithms; CSCI 211: Computer Programming II

Summer Engineering Academies (SEA) Staff Member

May 2019 – August 2019

University of Alaska Anchorage

- Facilitated the activities and learning of programming and robotics camps consisting of 20-30 kids from grades ranging from fourth to twelfth grade.

SKILLS

Programming Languages: C/C++, Python, Java, R, Bash

Tools and libraries: Git, GitHub, Tensorflow, PyTorch, Numpy, Pandas, Matplotlib

VOLUNTEER SERVICE

CURVE Mentor

2024

University of Southern California

- Mentoring three undergraduate students working on differentially private in-context learning and prompt optimization.

CSRMP Alumni Panel Discussion

2022

Google